

Mehul Vishwakarma

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SUMMARY

B.Tech AI & ML (2027) building full-stack + ML systems with FastAPI, React, SQL, Scikit-learn and Gemini API. Built deployed recommender, AI health assistant, and OCR automation pipeline. Internship experience deploying ML with Streamlit. Strong in Python, OOP, DSA, DBMS.

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript

Frontend: React, Vite, Tailwind CSS, shadcn/ui

Backend & APIs: FastAPI, Flask, REST APIs

Databases: MySQL, PostgreSQL

Core ML & LLM: Scikit-learn (Random Forest, Regression, Trees), Gemini API (gemini-1.5-flash), Prompt Pipeline, Feature Engineering, Model Evaluation (R2, MAE, F1)

Data & Viz: Pandas, NumPy, Matplotlib

Tools & Deployment: Git, GitHub, VS Code, Jupyter, Streamlit, Render, Vercel, Netlify

CS Fundamentals: OOP, Data Structures & Algorithms, DBMS

PROJECTS

PathFinder | AI-Powered Personalized Learning Path Recommender

FastAPI | Python | SQL | Content-Based Filtering, Cosine Similarity | GitHub: github.com/mehulvinodv/pathfinder | Live: path-finder-zeta-blond.vercel.app/dashboard

- Built 7+ REST APIs for learner profiles, career paths, skills, prerequisites, and progress tracking on SQL backend.
- Engineered content-based filtering with cosine similarity to map skill gaps against career-roadmap matrix and rank missing skills by prerequisite order.
- Built resource-matching pipeline to return targeted learning resources per gap with coverage/latency tracking.

HealthAI Guardian | AI-Powered Health Assistant - VOID Hacks 7.0, Top 15 / 250+

React, TypeScript, Vite, Tailwind CSS, Gemini API (gemini-1.5-flash), Scikit-learn | GitHub: [HealthAI-Guardian-void-hacks7.0](https://github.com/mehulvinodv/healthai-guardian) | Live: healthguardianai-two.vercel.app

- Built full-stack health assistant with React + Vite frontend for real-time symptom analysis and early risk prediction (Diabetes, Alzheimer's) using pretrained Scikit-learn models.
- Integrated Gemini 1.5 Flash via LLM prompt pipeline for conversational triage with speech-to-speech voice input, emergency symptom flagging, and auto-generated health reports.

FRA Atlas | Digitization & Decision Support Pipeline

Python | FastAPI | MySQL | Tesseract OCR | Rule-Based Engine | GitHub: github.com/mehulvinodv/fra-atlas

- Designed automated ingestion pipeline with Tesseract OCR to extract beneficiary fields from scans into MySQL.
- Added preprocessing + field validation before storage to handle noisy scans with field-level accuracy checks.
- Implemented rule-based decision engine over extracted fields to map beneficiaries to eligible schemes, cutting manual review.

EXPERIENCE

AI & Green Skills Intern | Edunet Foundation, AICTE & Shell - Jul 2025 - Aug 2025

AI Smart Irrigation System | Python, Scikit-learn (Random Forest), Pandas, Streamlit | Live: [smart-irrigation-app](https://smart-irrigation-app.vercel.app)

- Developed multi-output Random Forest regressor to jointly predict soil moisture / irrigation need from soil, temp, humidity, pH, and weather features.
- Ran preprocessing, feature engineering, train-test split, and hyperparameter tuning; evaluated with R2 / MAE. [Fill: R2 0.XX, MAE X.XX on N rows]
- Shipped Streamlit app for zone-wise real-time recommendations; scaled notebook prototype to deployment with IoT sensor data pipeline. [Add: trained on N records]

EDUCATION

Shri Ram Institute of Technology, Jabalpur - B.Tech CSE (AI & ML), 2023-2027

CGPA: 8.65

Central Academy, Shahdol - 12th (PCM), 2023: 91.8% | Spring Dales, Shahdol - 10th, 2022: 81.6%

ACHIEVEMENTS

- Hackathon Finalist (2x): NexGen Hack Ghaziabad 2025 - Top 30 / 400+ teams; VOID Hacks 7.0 - Top 15 / 250+ (HealthAI Guardian).